

DATA STORM 7.0 – TECHNICAL SUMMARY REPORT

DATA DYNAMOS : Latent Demand Estimation Framework

DELIVERABLES

- Data_Dynamos_predictions.csv – 11,121 outlets with uncapped demand predictions
- catboost_predictions.csv – Detailed prediction outputs
- model_clean.ipynb – Production-ready machine learning notebook

EXECUTIVE SUMMARY

Traditional sales forecasting methods rely heavily on historical sales averages. However, historical sales often fail to represent the true market demand because outlet performance can be constrained by stock shortages, delivery limitations, operational inefficiencies, and credit restrictions.

To address this challenge, the Data Dynamos team developed a **latent demand estimation framework** capable of predicting the realistic uncapped sales potential of beverage retail outlets across Sri Lanka.

The solution was built using a modern **Bronze → Silver → Gold medallion data architecture**, advanced feature engineering, geospatial enrichment, and machine learning techniques.

Key Results

- R² Score: **0.9943**
- MAE: **3.40 liters**
- Predictions generated for **11,121 outlets**
- 17 engineered features used for demand estimation

1: DATA FORENSICS & HYGIENE

1.1 System Anomalies Detection

Data quality validation was performed during the Silver layer transformation process.

Transaction-Level Anomalies

Anomaly Type	Detection Logic	Action
Negative sales volumes	Volume_Liters < 0	Quarantined
Invalid bill values	Total_Bill_Value < 0 OR NULL	Quarantined
Future-dated records	Year > 2025	Quarantined
Missing outlet IDs	gitOutlet_ID IS NULL	Quarantined
Duplicate transactions	Hash-based duplicate checks	Removed

Outlet-Level Validation

Validation	Result
Referential integrity checks	Validated against outlet_master
Invalid geographic coordinates	Removed
Duplicate outlet records	Deduplicated
Missing outlet attributes	Standardized

Seasonality Validation

- Missing months interpolated using forward-fill methods
- Extreme seasonality values capped at the 95th percentile
- Missing seasonality records filled using outlet averages

1.2 Rejected Records Quarantine System

Invalid records were isolated rather than permanently deleted.

Rejected Record Categories

- Negative sales values
- Invalid bill values
- Missing outlet identifiers
- Invalid coordinates outside Sri Lanka boundaries

Quarantine Storage Structure

data/silver/rejected/

```
|— invalid_transactions.csv  
|— invalid_outlets.csv  
|— invalid_coordinates.csv
```

Each quarantined record retained:

- Original row values
- Rejection reason
- Validation layer
- Recovery tracking metadata

1.2 Data Validation Framework

A 5-layer validation framework was implemented.

Layer 1 – Schema Validation

- Required columns verified
- Correct data types enforced
- Null validation for key fields

Layer 2 – Domain Validation

- Valid year ranges checked
- Valid month ranges checked

- Non-negative sales validation

Layer 3 – Referential Integrity

- Transaction outlet IDs validated
- SKU integrity validation
- Distributor integrity validation

Layer 4 – Statistical Validation

- Outlier detection using IQR methods
- Distribution consistency checks
- Temporal continuity validation

Layer 5 – Deduplication

- Transaction-level deduplication
- Outlet-level deduplication
- Holiday-level deduplication

Validation Summary

- Data retention rate: **99.36%**
- Referential integrity: **99.99%**
- Duplicate removal completed successfully

2: POI DATA ACQUISITION & SPATIAL ENRICHMENT

2.1 External Geospatial Signals

Data Sources

The project integrated external spatial context using:

- OpenStreetMap (OSM)
- OSMnx API
- Government geospatial datasets

POI Categories Used

POI Type	Purpose
Schools	Foot traffic estimation
Hospitals	Population density indicator
Bus stations	Transportation hub analysis
Shopping centers	Commercial activity signal

2.2 POI-to-Outlet Mapping

A radius-based catchment strategy was implemented.

Spatial Logic

For each outlet:

1. Extract outlet coordinates
2. Identify nearby POIs within 2 km
3. Calculate density scores
4. Generate contextual demand indicators

Engineered Spatial Features

- nearby_schools_count
- transport_proximity
- school_density
- geo_cluster

Example

Outlet: OUT_00001

- Nearby Schools: 12
- Nearby Hospitals: 3
- Nearby Transport Hubs: 2

Result: High surrounding activity correlated with stronger demand potential.

2.3 Spatial Feature Engineering

OSMnx Query Example

```
schools = ox.features_from_polygon(  
    sri_lanka_polygon,  
    tags={'amenity': 'school'}  
)
```

Additional Processing

- KMeans clustering used for geographic grouping
- Distance calculations performed using Haversine formula
- API rate limits managed with batch processing and caching

3: CAUSAL BASE LOGIC & UNCAPPED POTENTIAL

3.1 The Left-Censored Demand Problem

Observed sales do not always represent true demand because they are constrained by:

- Stock availability
- Delivery capacity
- Credit limitations
- Operational restrictions

Therefore:

Observed Sales = $\min(\text{True Demand}, \text{Operational Constraints})$

Traditional forecasting models often underestimate real demand because they learn from constrained historical sales.

3.2 Latent Demand Ceiling Estimation

95th Percentile Method

The project estimated demand ceilings using the 95th percentile of outlet sales.

Formula

Potential_Target = 95th Percentile of Volume_Liters per outlet

Why 95th Percentile?

Metric	Maximum Sales	95th Percentile
Outlier Sensitivity	High	Low
Stability	Low	High
Realism	Weak	Strong

This approach reduces the impact of extreme spikes while preserving realistic high-demand behavior.

3.3 Machine Learning Model

Algorithm Used

CatBoost Regressor

Why CatBoost?

- Handles tabular data efficiently
- Captures non-linear relationships
- Strong performance with minimal preprocessing
- Provides feature importance analysis

Model Configuration

```
CatBoostRegressor(  
    iterations=500,  
    learning_rate=0.05,  
    depth=6,  
    loss_function='RMSE',  
    random_state=42  
)
```

Features Used

- Lag features
- Rolling averages
- Seasonal encodings
- Geographic clusters
- Outlet-level aggregates

3.4 Model Performance

Metric	Train	Test
RMSE	7.0861	7.1881
MAE	3.3814	3.4006
R ² Score	0.9945	0.9943

Interpretation

- Model explains **99.43%** of sales variance
- Average prediction error is approximately **3.4 liters**
- Strong generalization observed on test data

4: FEATURE IMPORTANCE & INSIGHTS

4.1 Top Predictive Features

Rank	Feature	Importance
1	product_avg_volume	53.96%
2	outlet_avg_volume	26.77%
3	rolling_mean_3	8.07%
4	lag_1	5.49%
5	month_cos	2.33%

Key Insight

Product popularity and outlet historical strength contribute more than 80% of predictive power.

4.2 Prediction Output

Final Submission File

Data_Dynamos_predictions.csv

Output Columns

Column	Description
Outlet_ID	Unique outlet identifier
Maximum_Monthly_Liters	Predicted uncapped demand

Distribution Statistics

- Mean Potential: 572.04 liters/month
- Median Potential: 420.15 liters/month
- Maximum Potential: 5,741.22 liters

5: GENERATIVE AI TRANSPARENCY LOG

5.1 GenAI Tools Used

The following AI tools were used during the 36-hour competition sprint:

- ChatGPT
- GitHub Copilot
- Perplexity AI

5.2 Areas Where GenAI Was Used

Architecture Design

- Bronze -> Silver -> Gold pipeline planning
- Validation framework recommendations

Feature Engineering

- Lag feature implementation
- Rolling mean calculations
- Cyclical encoding logic

Machine Learning

- CatBoost parameter guidance
- Temporal train/test split recommendations

Documentation

- Technical writing support
- Report formatting assistance
- Grammar and structure refinement

5.3 Human Validation & Decision-Making

All final decisions were independently validated by the Data Dynamos team.

Team Responsibilities

- Business problem definition
- Model validation
- Feature interpretation
- Hyperparameter selection
- Final submission verification

GenAI Role

GenAI acted only as:

- Engineering accelerator
- Coding assistant
- Documentation helper

All business logic, validation, and model interpretation remained fully team-driven.

CONCLUSION

The Data Dynamos team successfully developed a complete latent demand estimation framework capable of predicting realistic uncapped beverage demand across Sri Lankan retail outlets.

The framework combined:

- Modern medallion data architecture
- Robust data engineering
- Advanced feature engineering
- Geospatial enrichment
- Machine learning prediction models

Final Achievements

- Successfully processed over 3.1 million records

- Engineered 17 meaningful predictive features
- Achieved 99.43% model accuracy (R^2)
- Generated predictions for 11,121 outlets
- Produced actionable business intelligence for FMCG distribution optimization

This solution enables smarter inventory allocation, improved outlet prioritization, and more effective business decision-making.